Purdue University – UNP NS-12

Title:	Payload Interface Control Document		
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Status	DRAFT		

Release Table

Action	Team Member	Role	Date
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Change Log

Version	Date	Prepared By	Description
1.0	9/30/2025	Nick Babusis	Initial Release
2.0	9/16/2026	Alan Hsu	CDR Submission

Overview

The APEx payload subsystem consists of the systems required to operate and characterize the PMPD thruster.

Table 1: Payload Specifications

Bounding Volume	180mm (X) x 88mm (Y) x 120mm (Z)
Mass	1.4kg (TBR)
Typical Power Draw	15W (thrusting; TBR), 1W (standby; TBR)
Power interfaces	Raw Battery 12-16.8V 5V (TBD) 3.3V 12V (EGSE)
Electrical Connectors	1x 26-pin Harwin Gecko (Main Power/Data) 1x 4-pin Harwin M80 (EGSE)
Mounting Connections	Up to 20x screws on +X/-X panels
Fluid Connections	2x 10-32 Flat-bottom ports on propellant tank
Communication Protocols	SPI, I2C, GPIO Pulse Signal
Microcontroller	STM32G431 (TBR)
Sensor Channels	1x Flow meter (Frit) 1x Pressure/Temperature Sensor (Frit) 2x Pressure/Temperature Sensor (Tank, 1 for refrigerant 1 for propellant) 2x Thermistors (Tank) 2x Thermistors (Payload Interface Board) 1x Ignitor Peak Voltage 1x Ignitor Integral Voltage 1x Ignitor Peak Current (Hall effect) 1x Ignitor Integral Current (Hall effect) 1x Ignitor Total Pulse Energy 1x Discharge Voltage 1x Discharge Current (Rogowski)
Actuators	2x Polyimide Heater (Tank) 3x Solenoid (Tank to Frit) 1x Ignitor Switch

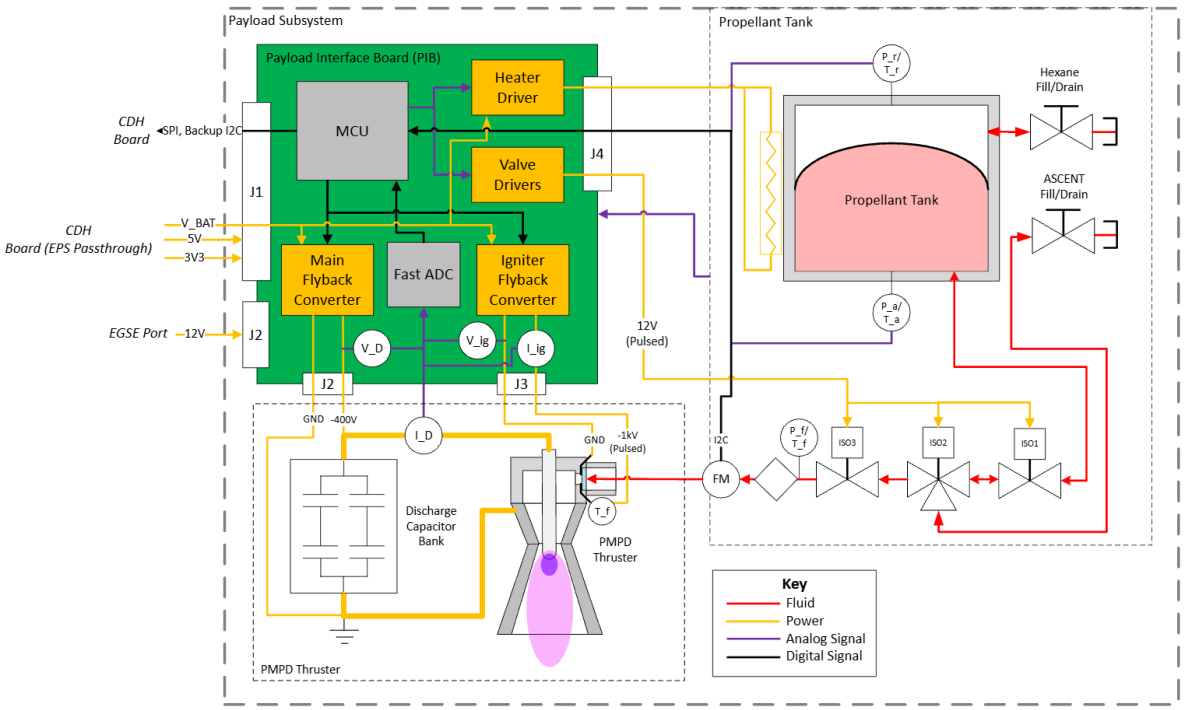


Figure 1: Full System Connection Diagram

Mechanical Interfaces

In terms of structural security, the payload subsystem is largely secured to the satellite via frame mounting, having multiple components within the subsystem mounted directly to said frame.

More specifically, the tank is mounted to an upper partition on the frame, which separates the payload from other subsystems. It is to be mounted via 4 screws on the top face of the panel extending into the inner walls of the tank, screwed in the +Z direction, as well as 2 screws on either side of the tank, screwed in the +Y/-Y direction. Said mounting methods are pictured below for reference.

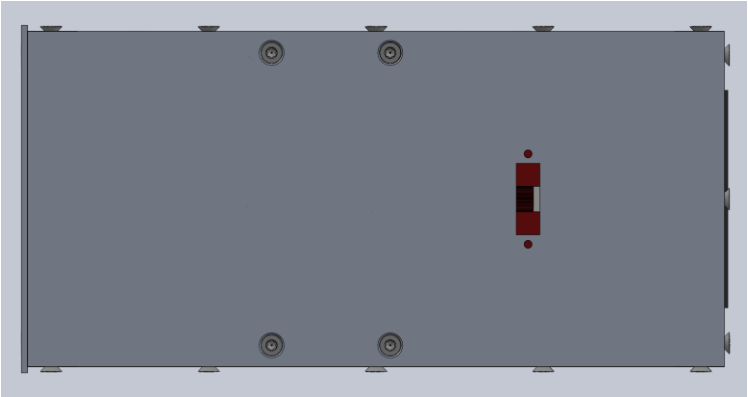


Figure 2: 4 Top Panel Tank Mounting Screws

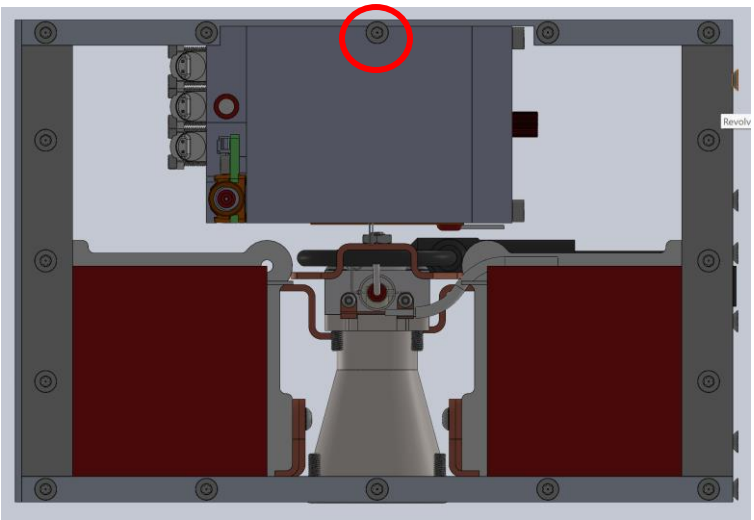


Figure 3: 2 Side Frame Tank Mounting Screws (Circled in Red & Mirrored on Other Side)

The capacitors are to be mounted to the frame as well, with the 2 of them being secured to the side panels of the frame, separating the inside of the satellite from the outside space. To be specific, each capacitor is secured via 4 screws on the nearest side panel, screwed in the +X/-X direction, Additionally, each capacitor has a bracket encasing it, which wrap around the capacitor and is secured to the side and bottom panels, via screwed in the +X/-X and –Z direction. Again, this system is pictured below for reference.

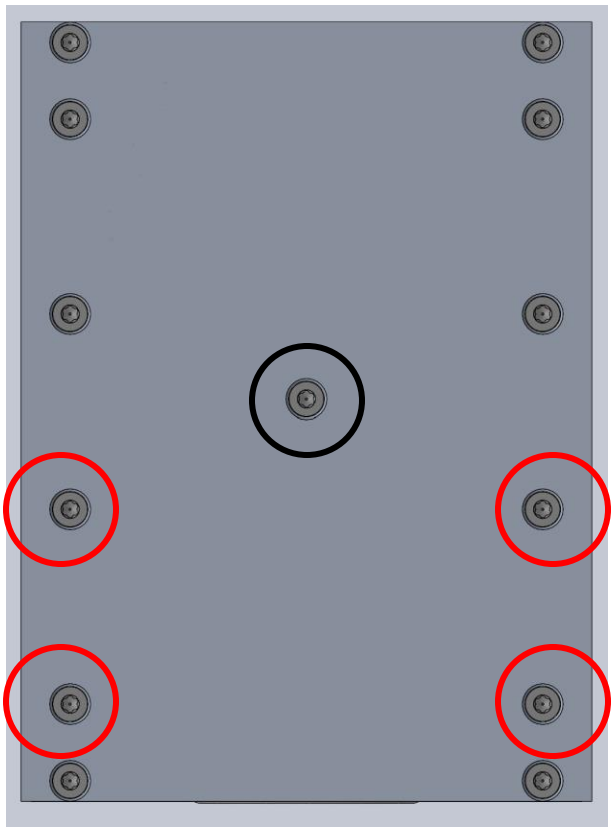


Figure 4: 2 Side Panel Capacitor Mounting and Bracket Screws (Capacitor Circled in Red, Bracket in Black & Mirrored on Other Side)

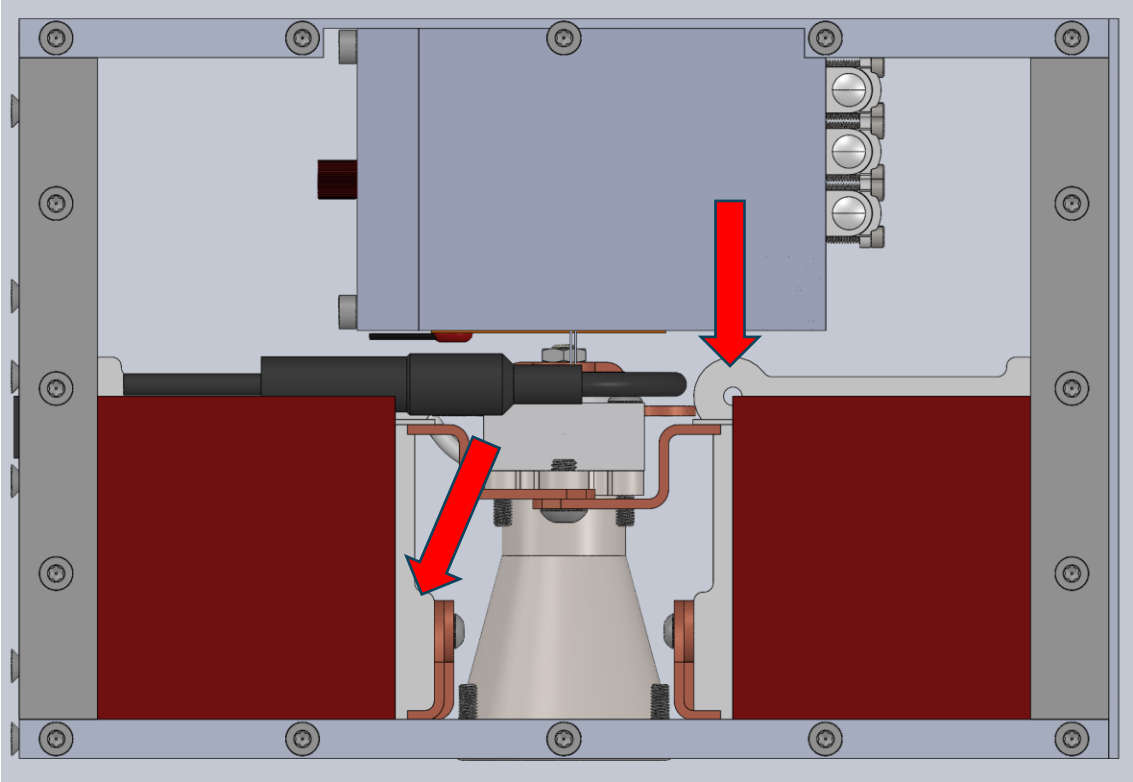


Figure 5: 2 Side View Capacitor Mounting and Bracket (Pointed to Via Red Arrow)

Additionally, the nozzle and accompanying frit are mounted to the capacitor securing brackets via screws in the +X/-X and +Z/-Z directions. That said, the nozzle specifically, is mounted to the bottom panel of the frame via 4 screws in the –Z direction, for extra security. Once more, images are provided for reference below.

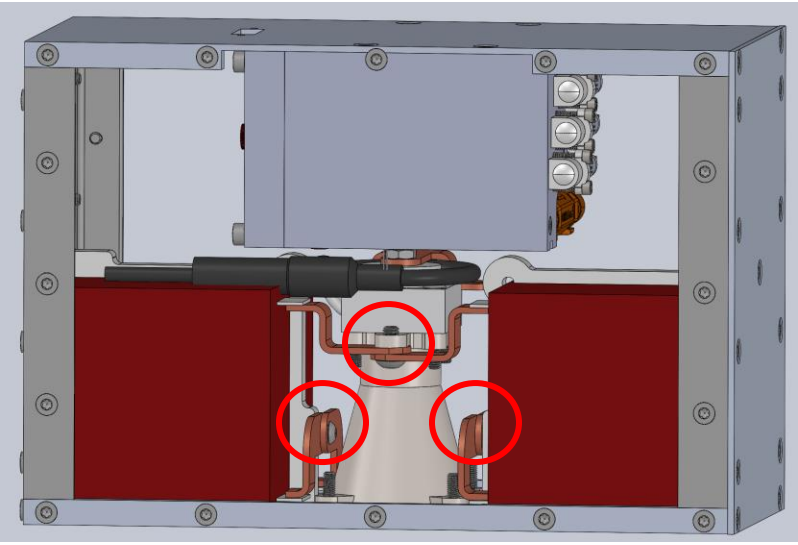


Figure 6: 2 Frit and Nozzle Bracket Securing (Circled in Red)

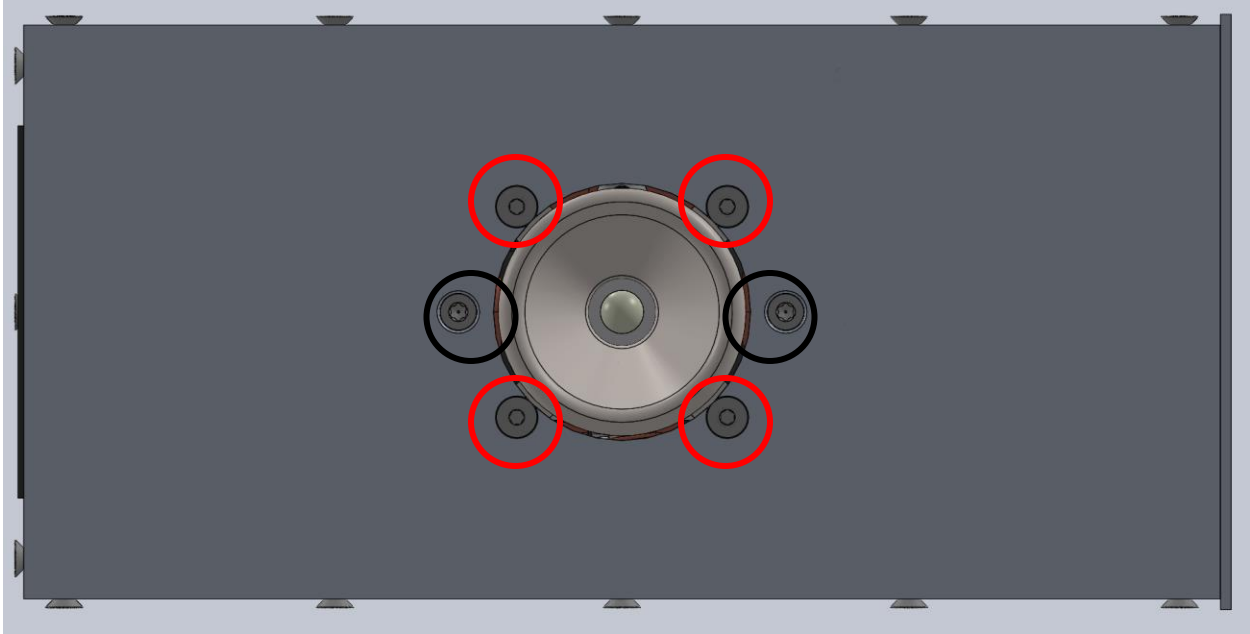


Figure 7: 2 Nozzle and Bracket Panel Securing Screws (Nozzle Circled in Red & Bracket in Black)

Electrical Interfaces

Pin	Func.	Pin	Func.
1	BAT	14	GND
2	BAT	15	GND
3	BAT	16	GND
4	3V3	17	GND
5	3V3	18	GND
6	SPI_MISO	19	GND
7	SPI_MOSI	20	I2C_SCL
8	SPI_SCK	21	I2C_SDA
9	SPI_CS	22	IGNITER
10	NC	23	DRDY
11	NC	24	ARM
12	5V	25	GND
13	5V	26	GND

Table 2: G125-MH12605M4P connector pinout

The Payload Interface Board (PIB) will expose a single G125 26 pin connector intended to interface with CDH. The G125-MH12605M4P connector has a current rating of 2.8A for an individual contact, and 2A per contact when multiple contacts are conducting. At 15W average and 12V (worst case) battery voltage three pins allocated to battery voltage means 0.42A per contact. This is ideal because the crimp of G125-MH12605M4P has a maximum wire gauge of 26 AWG, which is not well suited for high current. Redundancy is also achieved by using a minimum of two pins per power connection.

3.3V will be used for powering the microcontroller, digital logic, and some analog circuitry. 5V is claimed on the connector but TBR whether it will be used. Possible use cases include gate drive voltage for switching converters and/or a higher voltage to regulate a precise 3.3V analog reference voltage from. BAT will power the valve drivers through a 12V onboard regulator, discharge supply, and ignitor. The majority of the electrical power while thrusting will be delivered through the BAT bus, with only a few watts delivered for logic on the 3.3V and 5V buses.

TBR which bus will power heaters, depending on how much power is thermally required.

BAT will have overcurrent protection to ensure failures on PIB do not propagate to the upstream bus. 3.3V and 5V are assumed to have upstream protections provided by EPS. Additionally, onboard power circuitry will only pass BAT to the converters, valves, and heaters, if 3.3V is also powered and the microcontroller is healthy. Both HV converters are enabled/disabled using efuse ICs connecting/disconnecting the BAT bus with a MOSFET, independently of one another and of the other devices powered by BAT bus, ensuring fault isolation.

Data Interfaces

PIB exposes a primary SPI connection to the MCU, which acts as an SPI slave. A backup I2C interface is provided should SPI fail but acts at a lower bitrate. Three GPIO indicate when ignitor pulses happen at the hardware (IGNITOR), when ADC data is collected and ready for retrieval by CDH (DRDY), and whether the payload is ARMED or SAFE (ARM). All three are output only from the PIB MCU.

PIB firmware stores only an ARM/SAFE state, which determines whether high voltage supplies are active and whether propellant solenoid valves are open. ARM serves to ensure safety on potential ground testing, to ensure spurious commands do not cause thrust, and to minimize the voltage stress applied to high voltage converters on orbit by keeping them cold outside of thrusting periods. PIB defaults to SAFE and ARMs only upon the correct SPI/I2C command from CDH. Lack of communication from CDH for a specified watchdog interval will result in PIB defaulting back to SAFE state automatically, in case of comms failure. PIB polls all sensors except waveform sensors for telemetry data at 10 Hz (TBR) frequency, and data is stored in RAM and overwritten every polling cycle. Waveform data is recorded on a per pulse basis and is overwritten each pulse with the assumption that CDH will retrieve necessary data between pulses. The intent is for CDH to handle all complex states and reliable data storage, and for PIB to only actuate the thruster and immediately return the data to CDH.

Intended CONOPS is as follows:

1. Before experiment start, CDH issues command over SPI/I2C to ARM the payload. PIB responds by opening the three solenoid valves and enabling the high voltage discharge and ignitor supplies. After basic health checks come back OK, PIB asserts the ARM flag.
2. During the experiment, CDH will issue a command to pulse the thruster over SPI/I2C approximately once every second with contents including ignitor pulse width as an argument, and payload will respond by activating the ignitor upon

receipt of such command, and pulsing the IGNITOR signal with positive edge aligned with ignition.

3. Payload collects voltage/current waveform data from the pulse for 20us and asserts the DRDY flag to indicate to CDH that data is ready for retrieval.
4. CDH collects the data by SPI/I2C read command, requiring about 330 bytes per pulse. PIB de-asserts the DRDY flag upon data retrieval completion
    a. Data retrieval should take less than 1ms including overhead, assuming an SPI CLK frequency of 1MHz. This allows multiple attempts at data retrieval in case of failed transactions and ample time for non pulse related telemetry/command between pulses.
5. Repeat (2-4) until burn is complete, then CDH commands PIB to SAFE the payload, to which payload responds by switching off the high voltage rails and closing all valves, then de-asserting ARM
6. If SPI/I2C comms is lost mid-burn, software watchdog on PIB SAFEs the payload automatically after a set (on the order of 10s; TBR) timeout period without command.
7. Before, during, and after burn, CDH may issue commands for changing tank temperature setpoint, and PIB will regulate tank thermistor temperatures to these setpoints.
8. CDH may also telemeter flow rates, pressure/temperatures, and other housekeeping data not related to pulse waveforms at any point, regardless of DRDY flag. Retrieved data will be from the last time PIB polled sensors, which occurs at a TBR frequency loop.

See the Command & Telemetry List document for more details including a register map for commands accepted by PIB.

GSE Interfaces

Fill and drain of propellant and pressurant is done through 10-32 fittings on the propellant and pressurant sides of the tank. The +Y or –Y panel must be removed to access the tank within. 10-32 Flat-bottom ports on each side are nominally fitted with plugs, which can be removed and replaced with fittings to perform fill and drain.

Since fill/drain of propellant requires two of three solenoid valves to be open, separate circuitry on PIB opens these two specific valves upon application of 12V to the PIB EGSE connector. Both pairs of 12V input must be provided 12V for the valves to open. The circuitry driving the valves from the EGSE port's power is completely separated from spacecraft power and cold during flight, diode OR'd with the flight valve drivers to ensure

no interference. The connector pinout is listed below in detail and is passed through the satellite chassis EGSE connector.

Pin	Func.	Pin	Func.
1	12V_1	14	GND
2	12V_2	15	GND

Table 3: M80-5100442 connector pinout

Fill drain is designed to occur while the satellite is powered off, with EPS inhibited from powering the satellite.

Thermal Interfaces

The payload primarily dissipates 5W (TBR) of heat in the cathode of the thruster, and this is transferred out through a Boron Nitride substrate to the titanium anode (nozzle) of the thruster and the copper conductors to the main discharge capacitors, both of which dissipate into the +Z face of the frame.

The payload interface board's high voltage converters are expected to dissipate (TBR) watts of switching inefficiency, which dissipate heat to an Aluminum contoured heatsink plate on the -X face of the tank assembly, dissipating heat both into the tank and into the –Z panel of the PIB, which interfaces with the frame. This helps with maintaining the temperature of the propellant and pressurant in the tank.

The tank is maintained around 37˚C using polyamide heaters. The heaters are mounted on the +Z face of the tank assembly, and waste heat from the heaters radiates towards the +Z face and Boron Nitride substrate. The tank itself, made of Delrin plastic, is mounted to and dissipates heat to the –Z face of the frame.

TBD whether copper thermal straps are added to improve conduction to the frame. One risk item is the titanium selected for the anode, which has poor thermal conductivity compared to copper.

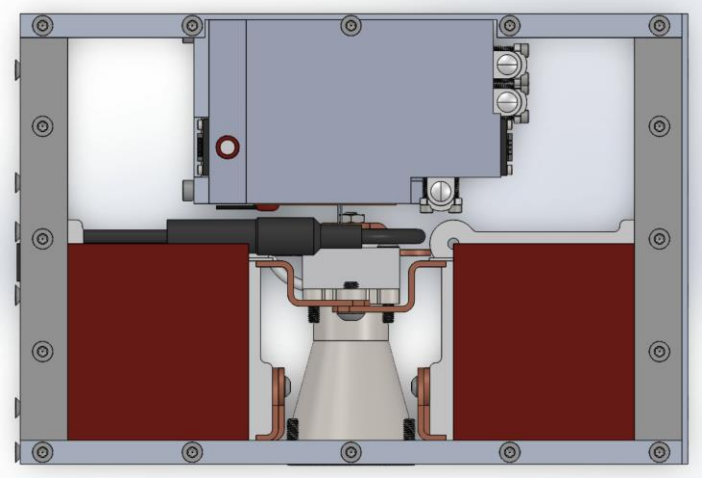


Figure 8: Payload Side View (TBR), note copper conductors connecting discharge capacitors and nozzle

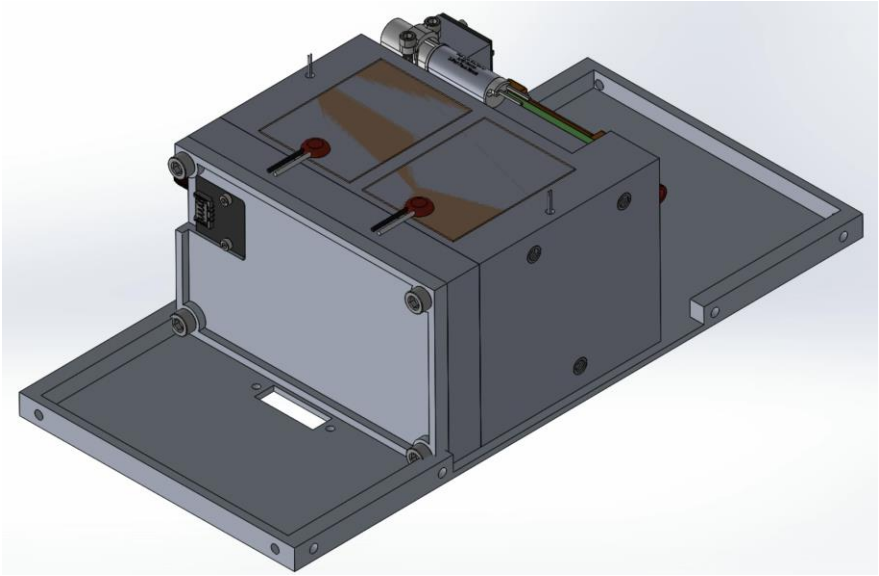


Figure 9: Tank assembly with –Z Panel